WORKED SOLUTION

STEP III 1998, Question 14

Probability and Statistics • Unbiased Estimators

STEP III 1998, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q14

This question has a full guided treatment in *Beyond STEP*, Section 13.3, *The German Tank Problem*: https://beyondstep.keithmcnulty.org/ch13/german-tanks.html

The key idea

Z_2 is a sum of independent pieces, so its mean and variance come straight from those of a single uniform variable. Z_1 depends on the maximum, and the natural way to handle a maximum is through its distribution function: the largest of the X_i is at most m exactly when *every* X_i is at most m. That turns into a product of probabilities, and differentiating gives the density.

Solution

The mean of Z_2

Each X_i has density $1/N$ on $[0, N]$, so

$$\mathrm{E}(X_i) = \frac{N}{2}, \qquad \mathrm{E}(X_i^2) = \frac{1}{N}\int_0^N x^2\,\mathrm{d}x = \frac{N^2}{3}, \qquad \mathrm{Var}(X_i) = \frac{N^2}{3} - \frac{N^2}{4} = \frac{N^2}{12}.$$

By linearity of expectation,

$$\mathrm{E}(Z_2) = \frac{2}{n}\sum_{i=1}^{n}\mathrm{E}(X_i) = \frac{2}{n}\cdot n\cdot\frac{N}{2} = N.$$

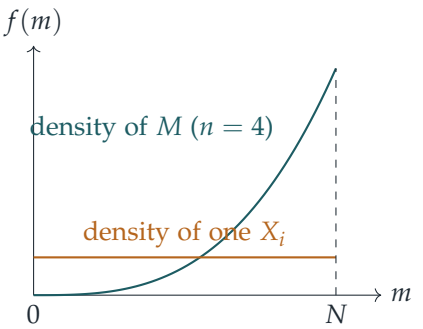
The distribution of the maximum

Let $M = \max\{X_1, \ldots, X_n\}$. For $0 \le m \le N$, the event $M \le m$ happens exactly when $X_i \le m$ for every i, and the X_i are independent, so

$$\mathrm{P}(M \le m) = \prod_{i=1}^{n}\mathrm{P}(X_i \le m) = \left(\frac{m}{N}\right)^n.$$

Differentiating, M has density

$$f(m) = \frac{n\,m^{n-1}}{N^n}, \qquad 0 \le m \le N.$$



The maximum piles up near N, which is why it needs only a small correction factor to become unbiased. Its moments are

$$\mathrm{E}(M) = \int_0^N m \cdot \frac{n\,m^{n-1}}{N^n}\,\mathrm{d}m = \frac{n}{N^n}\cdot\frac{N^{n+1}}{n+1} = \frac{n}{n+1}N,$$
$$\mathrm{E}(M^2) = \int_0^N m^2 \cdot \frac{n\,m^{n-1}}{N^n}\,\mathrm{d}m = \frac{n}{n+2}N^2.$$

The mean of Z_1

$$\mathrm{E}(Z_1) = \frac{n+1}{n}\,\mathrm{E}(M) = \frac{n+1}{n}\cdot\frac{n}{n+1}N = N.$$

So both estimators have mean N, as required.

The variances

For Z_1, first

$$\mathrm{Var}(M) = \frac{n}{n+2}N^2 - \frac{n^2}{(n+1)^2}N^2 = \frac{n\big[(n+1)^2 - n(n+2)\big]}{(n+2)(n+1)^2}N^2 = \frac{n}{(n+2)(n+1)^2}N^2,$$

since $(n+1)^2 - n(n+2) = 1$. Scaling by $\left(\frac{n+1}{n}\right)^2$,

$$\boxed{\mathrm{Var}(Z_1) = \frac{(n+1)^2}{n^2}\cdot\frac{n}{(n+2)(n+1)^2}N^2 = \frac{N^2}{n(n+2)}.}$$

For Z_2, the X_i are independent, so their variances add:

$$\boxed{\mathrm{Var}(Z_2) = \frac{4}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{4}{n^2}\cdot n\cdot\frac{N^2}{12} = \frac{N^2}{3n}.}$$

Ask yourself Which is smaller, and is it always the same one?

The ratio is

$$\frac{\mathrm{Var}(Z_1)}{\mathrm{Var}(Z_2)} = \frac{3n}{n(n+2)} = \frac{3}{n+2}.$$

This equals 1 when $n = 1$ (and then $Z_1 = Z_2 = 2X_1$, so the two estimators are actually the same), and is less than 1 for every $n \geq 2$.

Which to prefer

Both are unbiased, so we compare their spreads. We prefer Z_1:

- For $n \geq 2$ it has strictly smaller variance, and the advantage grows with n: $\mathrm{Var}(Z_1)$ falls like $1/n^2$, while $\mathrm{Var}(Z_2)$ falls only like $1/n$. With $n = 10$, for example, the standard deviation of Z_1 is about $0.09N$ against about $0.18N$ for Z_2.
- Z_1 is always at least the largest identification number seen, which N must be. Z_2 can come out smaller than a number we have actually observed (for example, with readings $1, 2, 30$ we get $Z_2 = 22$), giving an estimate we know to be impossible.

Checks

- When $n = 1$ both formulae give $N^2/3$, as they must since $Z_1 = Z_2 = 2X_1$, and $\mathrm{Var}(2X_1) = 4 \cdot N^2/12$.
- A simulation with $N = 10$ and 200 000 trials gave sample means within 0.01 of 10 for both estimators, and for $n = 4$ sample variances 4.16 and 8.33, against $\frac{100}{24} \approx 4.17$ and $\frac{100}{12} \approx 8.33$.

Common slips

- Trying to work with the density of the maximum directly. Go through $\mathrm{P}(M \leq m)$ first; the independence makes it a simple product.
- Forgetting to square the constant: $\mathrm{Var}(cY) = c^2 \mathrm{Var}(Y)$, so the factors are $\left(\frac{n+1}{n}\right)^2$ and $\frac{4}{n^2}$.

Beyond STEP

This is the classic "German tank problem", which Allied statisticians used in the Second World War to estimate German tank production from serial numbers, with remarkable accuracy. The maximum is a *sufficient statistic* here: once you know it, the rest of the sample tells you nothing more about N. The Lehmann–Scheffé theorem then shows that Z_1 has the smallest variance of all unbiased estimators. Sufficiency and

efficient estimators are treated in first-year statistics; Blitzstein and Hwang, *Introduction to Probability*, is a good place to start on order statistics.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.